TELECOM CUSTOMER CHURN ANALYSIS

Customer Retention & Business Performance Analysis Report

Tools Used: Python | Excel | PostgreSQL | Power BI
Dataset: IBM Telco Customer Churn Dataset

EXECUTIVE SUMMARY

Customer churn is a significant business challenge for telecommunications companies, as losing customers can affect recurring billing and long-term customer value. This project analyses customer churn patterns to identify customer segments associated with higher churn and areas where retention strategies may be prioritised.

The analysis was conducted using Python for data cleaning and preliminary exploration, PostgreSQL for structured business analysis, and Power BI for interactive visualisation and performance monitoring.

The dataset contains 7,043 customer records, of which 1,869 customers have churned, representing an overall churn rate of approximately 26.54%.

The analysis identifies several important patterns:

- Month-to-month contract customers have the highest churn rate at 42.71%.
- Customers with 0–12 months of tenure have the highest churn rate at 47.44%.
- Fiber optic customers have a churn rate of 41.89%.
- Customers paying through electronic checks have a churn rate of 45.29%.
- Customers without Online Security and Tech Support have a churn rate of 48.96%.
- Churned customers have higher average monthly charges and shorter average tenure than retained customers.

The Power BI dashboard provides an interactive overview of customer retention performance through key performance indicators, churn analysis charts and customer segmentation filters. The analysis provides insights that can support customer retention planning, service improvement and customer relationship management.

1. INTRODUCTION

1.1 Background

Customer retention is an important factor in the telecommunications industry, where companies depend on maintaining long-term customer relationships and recurring subscription payments.

Customer churn occurs when customers discontinue their services or subscriptions. A high churn rate can affect customer growth, recurring billing and business performance.

Understanding the characteristics and behaviour of customers who leave a service provider can help businesses identify customer segments that may require additional attention.

1.2 Problem Statement

Telecommunication companies face challenges in identifying customer groups that are more likely to discontinue their services. Without analysing customer behaviour, contract preferences, service usage and billing patterns, it becomes difficult to identify areas requiring retention-focused strategies.

This project addresses the problem by analysing customer information and identifying patterns associated with customer churn.

1.3 Project Objectives

- Calculate the overall customer churn rate.
- Analyse churn across different contract types and tenure groups.
- Examine the relationship between monthly charges and customer churn.
- Identify churn patterns across payment methods and internet services.
- Evaluate the relationship between additional services and customer retention.
- Analyse accumulated customer charges across different customer segments.
- Develop an interactive Power BI dashboard to monitor churn performance.
- Provide business recommendations based on the analytical findings.

1.4 Scope of the Analysis

The analysis focuses on customer demographics, subscription details, service preferences, payment methods, tenure and billing information.

The project covers data preparation, exploratory analysis, SQL-based business analysis, dashboard development and management recommendations.

2. DATASET OVERVIEW AND METHODOLOGY

2.1 Dataset Overview

The project uses the IBM Telco Customer Churn dataset, which contains customer-level information relating to telecommunications services, account details, billing and churn status.

The dataset consists of 7,043 customer records and includes variables such as:

Category	**Important Variables**
Customer Information	CustomerID, Gender, SeniorCitizen, Partner, Dependents

Contract Details	Tenure, Contract, PaperlessBilling
Internet Services	InternetService, OnlineSecurity, OnlineBackup, TechSupport
Payment Information	PaymentMethod, MonthlyCharges, TotalCharges
Customer Status	Churn

The dataset provides the information required to examine customer retention patterns across different customer segments.

2.2 Tools and Technologies Used

Python: Used for data inspection, data cleaning, missing-value treatment, duplicate checking and preliminary exploratory analysis.

Excel: Used as part of the data preparation and project workflow.

PostgreSQL: Used to perform structured SQL analysis, calculate customer churn rates, aggregate billing information and examine customer segments.

Power BI: Used to develop an interactive dashboard with key performance indicators, charts and slicers for monitoring customer churn.

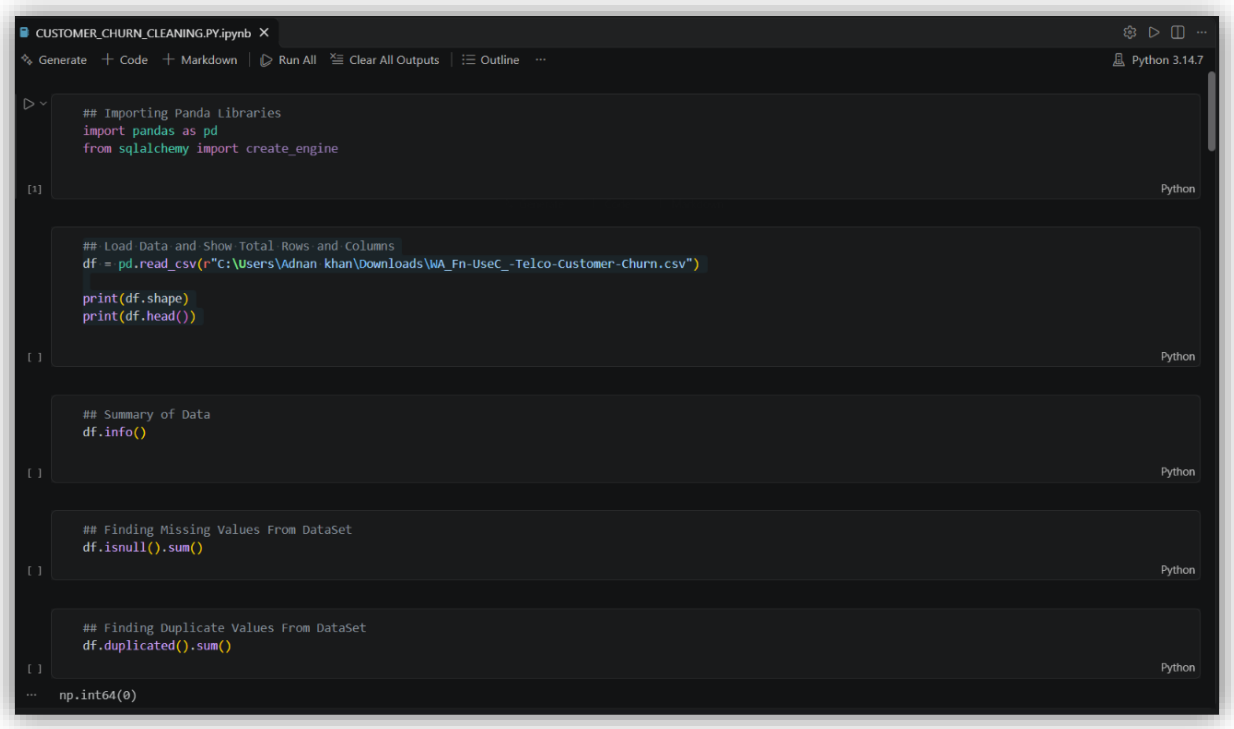
2.3 Data Cleaning and Preparation

Python was used to inspect and prepare the dataset before conducting the analysis.

The following data preparation activities were performed:

1. Inspected the dataset structure, dimensions and initial records.
2. Checked missing values and duplicate records.
3. Removed unnecessary leading and trailing spaces from text fields.
4. Identified inconsistent and whitespace-related values.
5. Converted the TotalCharges column into a numeric data type.
6. Identified 11 missing values in TotalCharges.
7. Verified that these missing values corresponded to customers with zero tenure.
8. Replaced the missing TotalCharges values with zero.
9. Examined descriptive statistics and checked for negative or unusual values.
10. Prepared the cleaned dataset for PostgreSQL analysis.

Figure 1: Jupyter Notebook Overview



Data Quality Results

Data Quality Check	Result
Total Customer Records	7,043
Duplicate Records	0
Missing TotalCharges Initially	11
Missing TotalCharges After Treatment	0
Negative Values Identified	None in the checks performed

The prepared dataset was imported into PostgreSQL for further analysis.

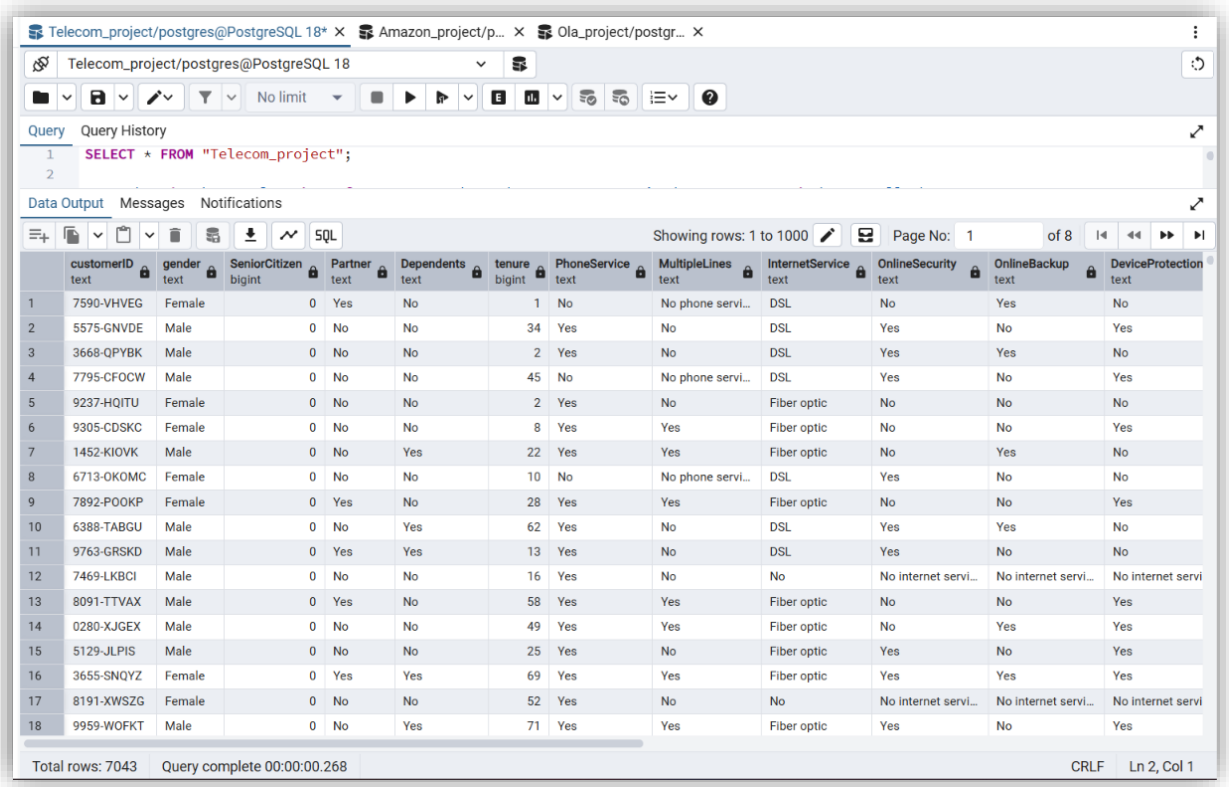
2.4 Analytical Methodology

The project followed a structured analytical workflow:

Data Collection → Data Inspection → Data Cleaning → Exploratory Analysis → SQL Analysis → Power BI Visualisation → Business Insights

Python was used for data preparation, PostgreSQL for detailed business analysis and Power BI for presenting the findings in an interactive format.

- Figure 2: PostgreSQL Table Overview



	customerID text	gender text	SeniorCitizen bigint	Partner text	Dependents text	tenure bigint	PhoneService text	MultipleLines text	InternetService text	OnlineSecurity text	OnlineBackup text	DeviceProtection text
1	7590-VHVEG	Female	0	Yes	No	1	No	No phone servi...	DSL	No	Yes	No
2	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes
3	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No
4	7795-CFOCW	Male	0	No	No	45	No	No phone servi...	DSL	Yes	No	Yes
5	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No
6	9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes
7	1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No
8	6713-OKOMC	Female	0	No	No	10	No	No phone servi...	DSL	Yes	No	No
9	7892-POOKP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	No	Yes
10	6388-TABGU	Male	0	No	Yes	62	Yes	No	DSL	Yes	Yes	No
11	9763-GRSKD	Male	0	Yes	Yes	13	Yes	No	DSL	Yes	No	No
12	7469-LKBCI	Male	0	No	No	16	Yes	No	No	No internet servi...	No internet servi...	No internet servi
13	8091-TTVAX	Male	0	Yes	No	58	Yes	Yes	Fiber optic	No	No	Yes
14	0280-XJGEX	Male	0	No	No	49	Yes	Yes	Fiber optic	No	Yes	Yes
15	5129-JLPIS	Male	0	No	No	25	Yes	No	Fiber optic	Yes	No	Yes
16	3655-SNQYZ	Female	0	Yes	Yes	69	Yes	Yes	Fiber optic	Yes	Yes	Yes
17	8191-XWSZG	Female	0	No	No	52	Yes	No	No	No internet servi...	No internet servi...	No internet servi
18	9959-WOFKT	Male	0	No	Yes	71	Yes	Yes	Fiber optic	Yes	No	Yes

3. CUSTOMER CHURN AND BUSINESS PERFORMANCE ANALYSIS

This section presents the findings from 15 SQL-based business analyses conducted using PostgreSQL.

3.1 Overall Customer Churn Analysis

The initial analysis examined the total number of customers, churned customers, retained customers and the overall churn rate.

Customer Status	Number of Customers
Total Customers	7,043
Churned Customers	1,869
Retained Customers	5,174
Churn Rate	26.54%

Key Finding: Approximately 26.54% of customers have discontinued their services, while 73.46% remain with the company.

This indicates that customer churn represents a significant area requiring attention from the company's retention management function.

3.2 Customer Churn by Contract Type

This analysis examined customer churn across month-to-month, one-year and two-year contract categories.

Contract Type	Total Customers	Churned	Retained	Churn Rate
Month-to-Month	3,875	1,655	2,220	42.71%
One Year	1,473	166	1,307	11.27%
Two Year	1,695	48	1,647	2.83%

Key Finding: Month-to-month contracts have the highest churn rate at 42.71%, whereas two-year contracts have the lowest at 2.83%.

Customers with longer contractual commitments demonstrate substantially lower churn rates than customers on month-to-month plans.

3.3 Customer Churn by Tenure Group

Customer tenure was divided into four groups to understand how the length of the customer relationship relates to churn.

Tenure Group	Total Customers	Churned	Churn Rate
0–12 Months	2,186	1,037	47.44%
13–24 Months	1,024	294	28.71%
25–48 Months	1,594	325	20.39%
49–72 Months	2,239	213	9.51%

Key Finding: Customers within their first 12 months have the highest churn rate at 47.44%. The churn rate progressively declines as customer tenure increases.

This highlights the importance of early-stage customer retention and effective onboarding strategies.

3.4 Customer Churn by Internet Service Type

This analysis compared churn rates among customers using Fiber Optic, DSL and customers without internet service.

Internet Service	Total Customers	Churned	Churn Rate
Fiber Optic	3,096	1,297	41.89%
DSL	2,421	459	18.96%
No Internet Service	1,526	113	7.40%

Key Finding: Fiber optic customers have the highest churn rate at 41.89%, followed by DSL customers at 18.96%.

The higher churn rate among fibre optic customers suggests that service experience, pricing and customer expectations within this segment may require further investigation.

3.5 Customer Churn by Monthly Charges

Customers were classified into low, medium and high monthly charge categories to examine differences in churn behaviour.

Monthly Charge Category	Total Customers	Churned	Churn Rate
Low	1,731	188	10.86%
Medium	1,721	407	23.65%
High	3,591	1,274	35.48%

Key Finding: Customers in the high monthly charge category have the highest churn rate at 35.48%, compared with 10.86% among low-charge customers.

This suggests an association between higher monthly charges and customer churn, highlighting the importance of reviewing pricing and perceived service value.

3.6 Customer Churn by Payment Method

This analysis examined customer churn across different payment methods.

Payment Method	Total Customers	Churned	Churn Rate
Electronic Check	2,365	1,071	45.29%
Mailed Check	1,612	308	19.11%
Bank Transfer (Automatic)	1,544	258	16.71%
Credit Card (Automatic)	1,522	232	15.24%

Key Finding: Electronic check users have the highest churn rate at 45.29%, while customers using automatic credit card payments have the lowest rate at 15.24%.

The difference indicates that payment method is an important customer segmentation variable for further retention analysis.

3.7 Customer Churn by Demographic Segments

This analysis examined customer churn across gender and senior citizen categories.

Customer Segment	Total Customers	Churned	Churn Rate
Female Senior Citizens	568	240	42.25%
Male Senior Citizens	574	236	41.11%
Female Non-Senior Citizens	2,920	699	23.94%
Male Non-Senior Citizens	2,981	694	23.28%

Key Finding: Senior citizen customers have higher churn rates than non-senior customers in both gender categories.

The difference between male and female churn rates is relatively small. Therefore, senior citizen status appears more relevant to this segmentation than gender.

3.8 Customer Churn by Online Security and Tech Support

This analysis examined churn across combinations of Online Security and Tech Support subscriptions.

Online Security	Tech Support	Total Customers	Churned	Churn Rate
Yes	Yes	1,099	99	9.01%
No Internet Service	No Internet Service	1,526	113	7.40%
No	No	2,553	1,250	48.96%
No	Yes	945	211	22.33%
Yes	No	920	196	21.30%

Key Finding: Customers without both Online Security and Tech Support have the highest churn rate at 48.96%.

Customers subscribed to both services have a substantially lower churn rate of 9.01%.

This indicates that additional service adoption is associated with differences in customer retention.

3.9 Comparison of Average Charges and Tenure by Customer Status

This analysis compared average monthly charges, tenure and accumulated customer charges between retained and churned customers.

Customer Status	Customers	Avg. Monthly Charges	Avg. Tenure (Months)	Avg. Total Charges
Retained	5,174	61.27	37.57	2,549.91
Churned	1,869	74.44	17.98	1,531.80

Key Finding: Churned customers have higher average monthly charges of 74.44 compared with 61.27 among retained customers.

Additionally, churned customers have an average tenure of only 17.98 months, compared with 37.57 months for retained customers.

The results indicate that churned customers tend to have shorter customer relationships and higher average monthly charges.

3.10 Customer Churn by Contract Type and Tech Support

This analysis examined whether churn patterns differed across contract categories based on Tech Support subscription status.

Contract Type	Tech Support	Total Customers	Churned	Churn Rate
Month-to-Month	No	2,680	1,350	50.37%
Month-to-Month	Yes	671	206	30.70%
Month-to-Month	No Internet Service	524	99	18.89%
One Year	No Internet Service	364	9	2.47%
One Year	No	557	82	14.72%
One Year	Yes	552	75	13.59%
Two Year	No	236	14	5.93%
Two Year	Yes	821	29	3.53%
Two Year	No Internet Service	638	5	0.78%

Key Finding: Month-to-month customers without Tech Support have the highest churn rate at 50.37%.

Within both one-year and two-year contract categories, the churn rate is lower than that of month-to-month customers.

The findings suggest that contract commitment and additional service subscriptions are useful variables for identifying higher-risk customer segments.

3.11 Accumulated Customer Charges by Contract Type

This analysis examined the total accumulated customer charges across different contract categories.

Contract Type	Total Customers	Total Customer Charges	Contribution
Two Year	1,695	6,283,253.70	39.13%
Month-to-Month	3,875	5,305,861.50	33.05%
One Year	1,473	4,467,053.50	27.82%

Key Finding: Two-year contract customers contribute approximately 39.13% of total accumulated customer charges, despite representing fewer customers than the month-to-month category.

This reflects the importance of longer customer relationships in accumulated customer billing value.

3.12 Accumulated Customer Charges by Payment Method

This analysis compared total accumulated charges, average accumulated charges and contribution across payment methods.

Payment Method	Customers	Total Charges	Avg. Total Charges	Contribution
Electronic Check	2,365	4,944,903.25	2,090.87	30.80%
Bank Transfer (Automatic)	1,544	4,748,279.90	3,075.31	29.57%
Credit Card (Automatic)	1,522	4,671,593.35	3,069.38	29.10%
Mailed Check	1,612	1,691,392.20	1,049.25	10.53%

Key Finding: Electronic check users account for the highest contribution to total accumulated customer charges at 30.80%.

However, automatic bank transfer and credit card users have higher average accumulated charges per customer than electronic check users.

3.13 Customer Churn by Online Security, Online Backup and Device Protection

This analysis examined customer churn across combinations of three additional services: Online Security, Online Backup and Device Protection.

Online Security	Online Backup	Device Protection	Total Customers	Churned	Churn Rate
No	No	No	1,509	793	52.55%
No	No	Yes	686	264	38.48%
No	Yes	No	678	236	34.81%
No	Yes	Yes	625	168	26.88%
Yes	No	No	475	118	24.84%
Yes	Yes	No	433	64	14.78%
Yes	No	Yes	418	58	13.88%
Yes	Yes	Yes	693	55	7.94%
No Internet Service	No Internet Service	No Internet Service	1,526	113	7.40%

Key Finding: Customers without all three additional services have the highest churn rate at 52.55%.

Customers subscribed to all three services have a churn rate of 7.94%.

The results indicate that customers with a broader combination of additional services demonstrate lower churn rates in this dataset.

3.14 Customer Segmentation by Accumulated Customer Charges

Customers were divided into three segments based on their TotalCharges values to understand the distribution of accumulated customer charges.

Customer Charge Segment	Customers	Total Charges	Average Charges	Contribution
Low	2,904	1,037,347.40	357.21	6.46%
Medium	1,935	3,458,593.80	1,787.39	21.54%
High	2,204	11,560,227.50	5,245.11	72.00%

Key Finding: The high accumulated-charge segment contributes 72% of total accumulated customer charges, despite accounting for approximately 31.3% of customers.

This segment represents a substantial share of the customer billing value accumulated over the recorded customer relationships.

3.15 Customer Churn by Accumulated Customer Charge Segment

This analysis examined churn rates across low, medium and high accumulated-charge customer segments.

Customer Charge Segment	Total Customers	Churned	Churn Rate
Low	2,904	1,070	36.85%
Medium	1,935	449	23.20%
High	2,204	350	15.88%

Key Finding: Customers in the low accumulated-charge segment have the highest churn rate at 36.85%, while the high segment has the lowest rate at 15.88%.

This pattern is consistent with the tenure analysis, where customers with shorter relationships demonstrate higher churn.

4. POWER BI DASHBOARD AND PERFORMANCE MONITORING

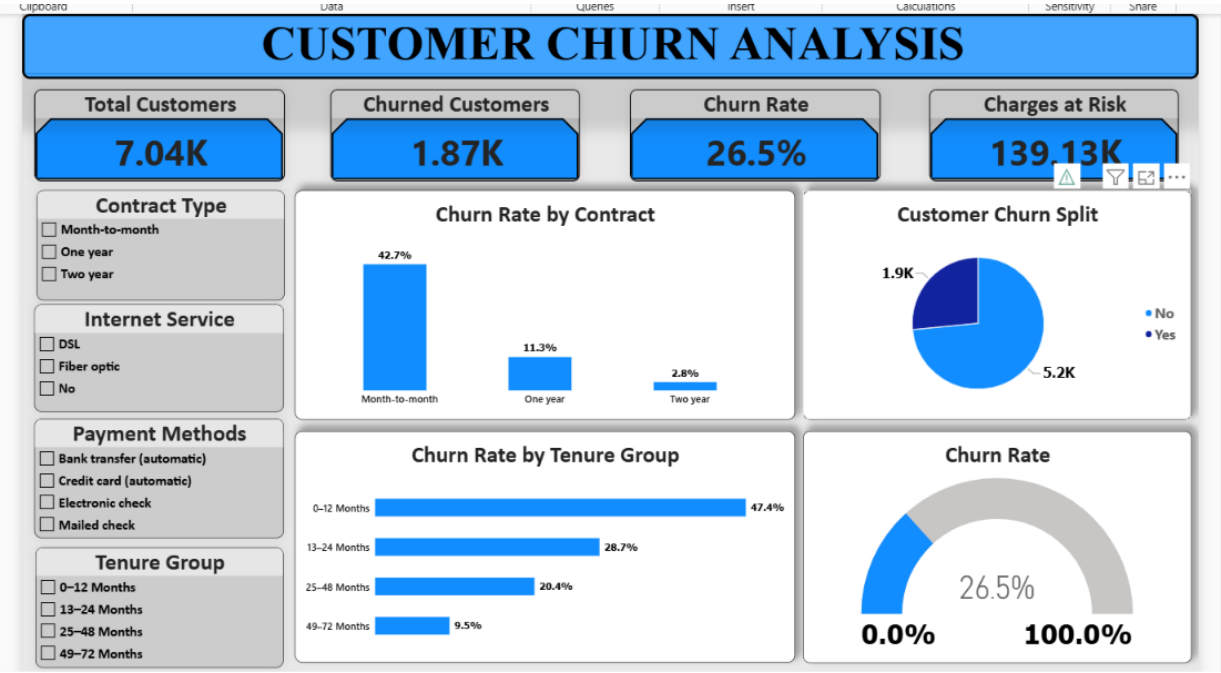
4.1 Dashboard Overview

An interactive Power BI dashboard was developed to provide a consolidated view of customer churn performance.

The dashboard presents key customer metrics and visual comparisons across contract types, tenure groups and customer churn status.

It enables users to examine customer retention patterns through interactive filters and visual summaries.

Figure 3: Complete Power BI Dashboard Overview



4.2 Key Performance Indicators (KPIs)

The dashboard includes four primary KPIs.

KPI	Value	Description
Total Customers	7.04K	Total number of customers in the dataset
Churned Customers	1.87K	Number of customers who discontinued services
Churn Rate	26.5%	Percentage of customers who churned
Monthly Charges at Risk	139.13K	Combined monthly charges associated with churned customers

Monthly Charges at Risk represents the sum of MonthlyCharges for customers whose Churn status is Yes. It indicates the monthly billing value associated with churned customers and should not be interpreted as confirmed annual revenue loss or profit loss.

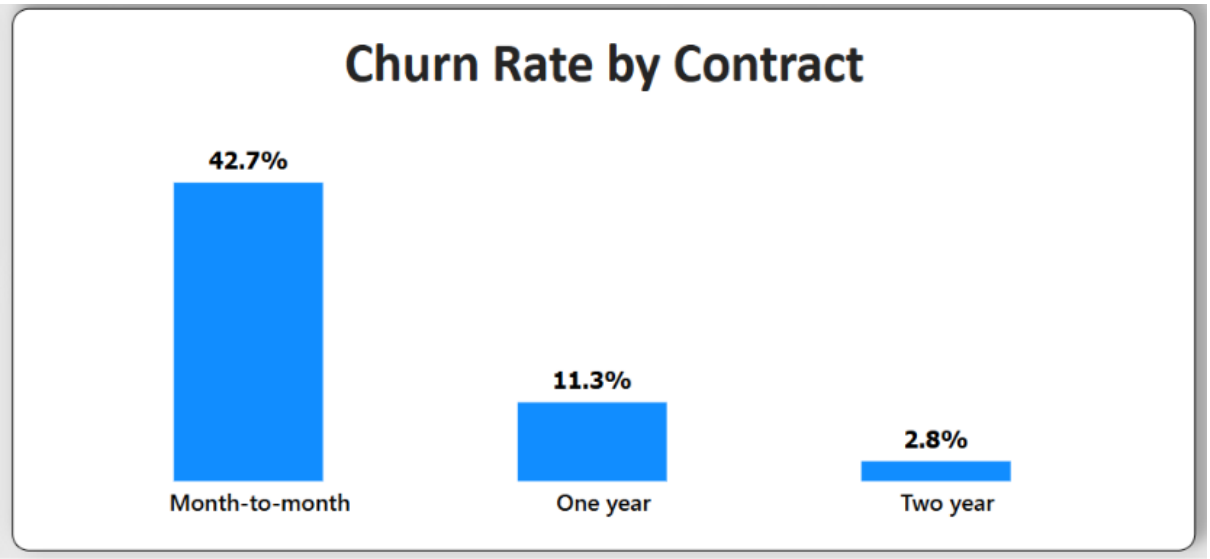
4.3 Dashboard Visualisations

The dashboard contains the following visualisations:

1. Churn Rate by Contract

A stacked bar visual comparing customer churn across different contract categories. It helps identify contract groups with higher customer churn.

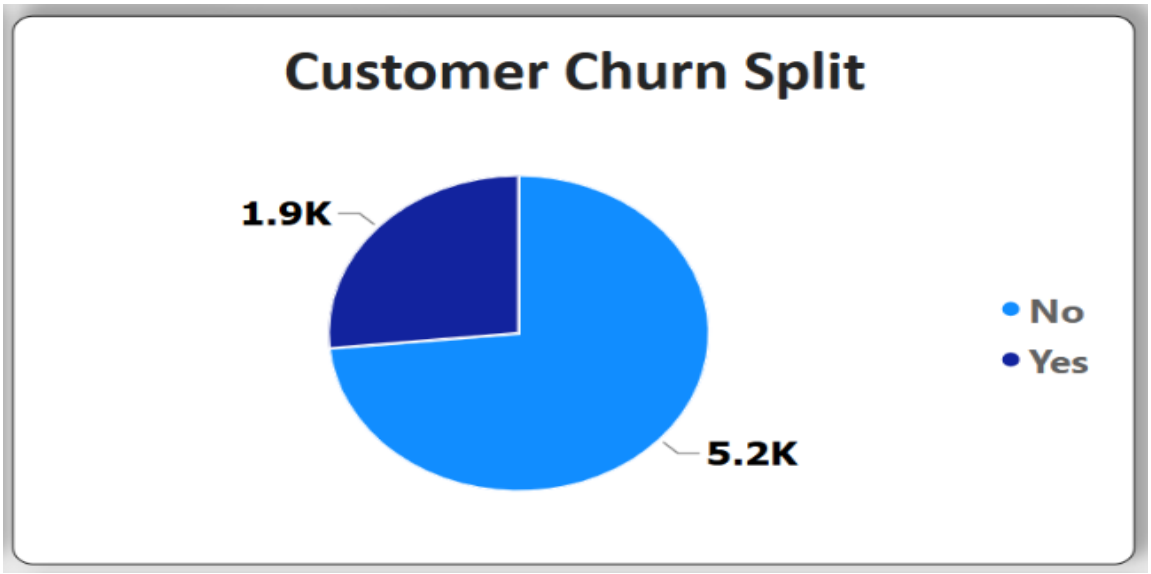
Figure 4: Churn Rate by Contract Type



2. Customer Churn Split

A pie chart showing the distribution of churned and retained customers, providing an overview of customer retention status.

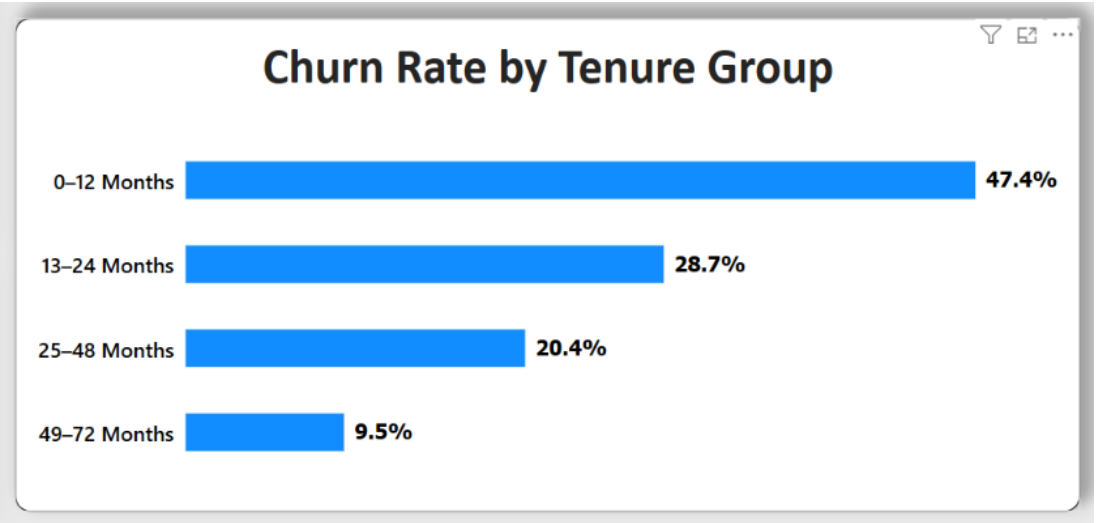
Figure 5: Customer Churn Split



3. Churn Rate by Tenure Group

A clustered bar visual comparing churn rates across different customer tenure categories. It highlights the higher churn observed among newer customers.

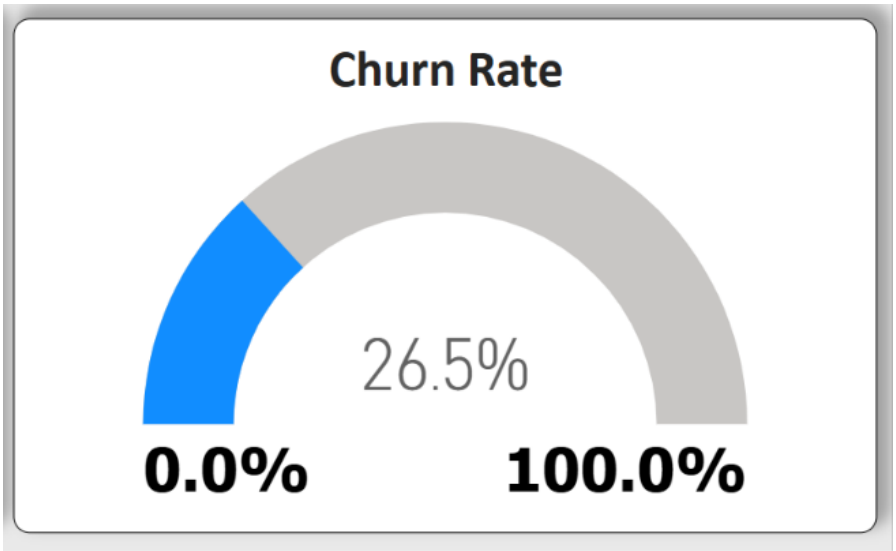
Figure 6: Churn Rate by Tenure Group



4. Churn Rate Gauge

A gauge visual displaying the overall customer churn rate of approximately 26.5% as a key retention performance indicator.

Figure 7: Churn Rate Gauge



4.4 Dashboard Filters and Interactivity

The dashboard includes four slicers to enable customer segmentation and interactive analysis:

- **Contract Type:** Filters customer information by contract category.
- **Internet Services:** Allows comparison across internet service categories.
- **Payment Methods:** Enables analysis of churn across different payment methods.

- **Tenure Group:** Filters customers according to their relationship duration.

These filters allow users to examine churn patterns across different customer segments without requiring separate reports for each category.

5. KEY BUSINESS INSIGHTS AND MANAGEMENT RECOMMENDATIONS

5.1 Early-Stage Customer Retention

Insight: Customers with 0–12 months of tenure have the highest churn rate at 47.44%.

Recommendation: Strengthen customer onboarding, provide early-stage assistance and monitor customer satisfaction during the initial months of service.

5.2 Month-to-Month Contract Retention

Insight: Month-to-month customers have a churn rate of 42.71%, compared with only 2.83% among two-year contract customers.

Recommendation: Introduce suitable long-term contract incentives and communicate the benefits of longer subscription plans to eligible customers.

5.3 Review of Fiber Optic Customer Experience

Insight: Fiber optic customers have the highest churn rate among internet service categories at 41.89%.

Recommendation: Examine service quality, pricing, connectivity issues and customer feedback within the fibre optic segment to identify potential areas for improvement.

5.4 Pricing and Customer Value

Insight: Customers in the high monthly charge category have a churn rate of 35.48%. Churned customers also have higher average monthly charges than retained customers.

Recommendation: Review pricing structures, identify customers who may perceive their plans as expensive and consider suitable service packages based on customer requirements.

5.5 Payment Method Improvement

Insight: Electronic check users have the highest churn rate at 45.29%, while automatic credit card users have the lowest at 15.24%.

Recommendation: Evaluate payment-related customer experiences and encourage convenient automatic payment options where appropriate.

5.6 Additional Service Adoption

Insight: Customers without Online Security and Tech Support have a churn rate of 48.96%. Customers subscribed to all three of Online Security, Online Backup and Device Protection have a churn rate of 7.94%.

Recommendation: Improve awareness of additional services and communicate their benefits through relevant customer engagement and subscription offers.

5.7 Senior Citizen Customer Retention

Insight: Senior citizen customers have churn rates above 41% across both gender categories.

Recommendation: Consider accessible customer support, simplified service plans and targeted engagement for senior citizen customers.

5.8 Protection of High-Value Customer Relationships

Insight: The high accumulated-charge customer segment contributes 72% of total accumulated customer charges.

Recommendation: Monitor high-value customer relationships and identify service or billing issues that could affect their continued subscriptions.

6. CONCLUSION AND ANALYTICAL LIMITATIONS

6.1 Conclusion

The Telecom Customer Churn Analysis project examined customer retention patterns using Python, PostgreSQL and Power BI.

The analysis of 7,043 customer records identified an overall churn rate of 26.54%, with significant differences across contract types, tenure groups, payment methods, internet services and additional service subscriptions.

The findings highlight that early-tenure customers, month-to-month subscribers, fiber optic customers and electronic check users represent important segments for further retention analysis.

The Power BI dashboard provides a consolidated and interactive view of customer churn performance through key metrics, visualisations and customer segmentation filters.

Overall, the project demonstrates how data analytics can transform customer information into meaningful business insights and support more informed customer retention decisions.

6.2 Analytical Limitations

- The analysis is based on the customer records available in the supplied dataset and may not represent the entire telecommunications market.
- The dataset provides a snapshot of customer information and does not establish the exact timing or sequence of customer churn events.
- The analysis identifies associations between customer characteristics and churn but does not establish direct cause-and-effect relationships.
- TotalCharges represents accumulated customer charges in the dataset and should not be treated as recognised revenue for a specific reporting period.
- Monthly Charges at Risk represents the combined MonthlyCharges of churned customers and is not a verified measure of actual financial loss.
- Customer segmentation results depend on the category thresholds used in the SQL analysis.